

## SAFETY DATA SHEET

Creation Date 24-April-2009

Revision Date 25-December-2021

Revision Number 9

### 1. Identification

**Product Name** Ethanol, 200 proof

**Cat No. :** AC397690000; AC397690010; AC397690025; AC397691000

**CAS-No** 64-17-5  
**Synonyms** Ethyl alcohol; Absolute ethanol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |            |
|------------------------------------------|------------|
| <b>Flammable liquids</b>                 | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 2 |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Highly flammable liquid and vapor  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Obtain special instructions before use

Do not handle until all safety precautions have been read and understood

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Use explosion-proof electrical/ventilating/lighting/equipment

Use only non-sparking tools

Take precautionary measures against static discharges

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

IF exposed or concerned: Get medical advice/attention

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up

Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component     | CAS-No  | Weight % |
|---------------|---------|----------|
| Ethyl alcohol | 64-17-5 | 99-100   |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms/effects**

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | Water may be ineffective, Do not use a solid water stream as it may scatter and spread fire                                             |
| <b>Flash Point</b>                      | 12 °C / 53.6 °F                                                                                                                         |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 363 °C / 685.4 °F                                                                                                                       |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 19 vol %                                                                                                                                |
| <b>Lower</b>                            | 3.3 vol %                                                                                                                               |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
3

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not flush into surface water or sanitary sewer system.                                                                        |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from open flames, hot surfaces and sources of ignition. Flammables area. Keep away from heat, sparks and flame. Incompatible Materials. Strong oxidizing agents. Strong acids. Acid anhydrides. Acid chlorides.                                                                                                                                  |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component     | Alberta                                      | British Columbia | Ontario TWAEV  | Quebec         | ACGIH TLV      | OSHA PEL                                                                                                         | NIOSH IDLH                                                     |
|---------------|----------------------------------------------|------------------|----------------|----------------|----------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Ethyl alcohol | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup> | STEL: 1000 ppm   | STEL: 1000 ppm | STEL: 1000 ppm | STEL: 1000 ppm | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> | IDLH: 3300 ppm<br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness   | Glove comments                                         |
|----------------|-------------------|-------------------|--------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.38 mm - 0.56 mm | As tested under EN374-3                                |
| Neoprene       | > 480 minutes     | 0.45 mm           | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                |                          |
|----------------|--------------------------|
| Physical State | Liquid                   |
| Appearance     | Clear, Colorless         |
| Odor           | Alcohol                  |
| Odor Threshold | No information available |
| pH             | 7 @ 20°C 10g/l aq.sol    |

|                                        |                                                       |
|----------------------------------------|-------------------------------------------------------|
| Melting Point/Range                    | -114 °C / -173.2 °F                                   |
| Boiling Point/Range                    | 78 °C / 172.4 °F                                      |
| Flash Point                            | 12 °C / 53.6 °F                                       |
| Evaporation Rate                       | No information available                              |
| Flammability (solid,gas)               | Not applicable                                        |
| Flammability or explosive limits       |                                                       |
| Upper                                  | 19 vol %                                              |
| Lower                                  | 3.3 vol %                                             |
| Vapor Pressure                         | No information available                              |
| Vapor Density                          | No information available                              |
| Specific Gravity                       | 0.785 g/cm3 @20°C                                     |
| Solubility                             | miscible                                              |
| Partition coefficient; n-octanol/water | No data available                                     |
| Autoignition Temperature               | 363 °C / 685.4 °F                                     |
| Decomposition Temperature              | No information available                              |
| Viscosity                              | No information available                              |
| Molecular Formula                      | C2 H6 O                                               |
| Molecular Weight                       | 46.07                                                 |
| VOC Content(%)                         | 100% (Organic Carbon (by mass) = 52.1 %) (EC/1999/13) |

## 10. Stability and reactivity

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                        |
| Stability                        | Hygroscopic.                                                                                                      |
| Conditions to Avoid              | Incompatible products. Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Acid anhydrides, Acid chlorides                                            |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                                           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                          |
| Hazardous Reactions              | None under normal processing.                                                                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component     | LD50 Oral                                                    | LD50 Dermal | LC50 Inhalation                                                   |
|---------------|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| Ethyl alcohol | LD50 = 10470 mg/kg<br>OCED 401 (Rat)<br>3450 mg/kg ( Mouse ) | Not listed  | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes                                                                       |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component     | CAS-No  | IARC       | NTP   | ACGIH | OSHA       | Mexico |
|---------------|---------|------------|-------|-------|------------|--------|
| Ethyl alcohol | 64-17-5 | Not listed | Known | A3    | Not listed | A3     |

IARC (International Agency for Research on Cancer)

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Group 1 - Carcinogenic to Humans

NTP: (National Toxicity Program)

ACGIH: (American Conference of Governmental Industrial Hygienists)

OSHA: (Occupational Safety & Health Administration)

Mexico - Occupational Exposure Limits - Carcinogens

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

OSHA: (Occupational Safety & Health Administration)

X - Present

Mexico - Occupational Exposure Limits - Carcinogens

A1 - Confirmed Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Confirmed Animal Carcinogen

A4 - Not Classifiable as a Human Carcinogen

A5 - Not Suspected as a Human Carcinogen

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component     | Freshwater Algae                              | Freshwater Fish                                                  | Microtox                                                                                                                | Water Flea                                    |
|---------------|-----------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Ethyl alcohol | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | Photobacterium<br>phosphoreum: EC50 = 34634<br>mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 = 35470<br>mg/L/5 min | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component     | log Pow |
|---------------|---------|
| Ethyl alcohol | -0.32   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT**

UN-No UN1170  
 Proper Shipping Name ETHANOL  
 Hazard Class 3  
 Packing Group II

**TDG**

UN-No UN1170  
 Proper Shipping Name ETHANOL  
 Hazard Class 3  
 Packing Group II

**IATA**

UN-No UN1170  
 Proper Shipping Name ETHANOL  
 Hazard Class 3  
 Packing Group II

**IMDG/IMO**

UN-No UN1170  
 Proper Shipping Name ETHANOL  
 Hazard Class 3  
 Packing Group II

## 15. Regulatory information

**International Inventories**

| Component     | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Ethyl alcohol | 64-17-5 | X   | -    | X    | ACTIVE                                        | 200-578-6 | -      | -   |

| Component     | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------|---------|-------|----------|------|------|------|------|-------|-------|
| Ethyl alcohol | 64-17-5 | X     | KE-13217 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component     | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|---------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Ethyl alcohol | Part 5, Individual Substances Part 4 Substance       |                                                                            |                                           |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component     | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Ethyl alcohol | 64-17-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component     | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Ethyl alcohol | 64-17-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |

**16. Other information**

|                         |                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Regulatory Affairs<br>Acros Organics BVBA<br>Tel: 800-ACROS-01                                                                                                                                              |
| <b>Creation Date</b>    | 24-April-2009                                                                                                                                                                                               |
| <b>Revision Date</b>    | 25-December-2021                                                                                                                                                                                            |
| <b>Print Date</b>       | 25-December-2021                                                                                                                                                                                            |
| <b>Revision Summary</b> | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. SDS sections updated. 2. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**